

# prepare\_coords\_for\_fit.R

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| prepare_coords_for_fit | <i>Prepare Coordinate Data for Model Fitting</i> |
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## Description

Expands dataset-level coordinate data to the sample level by joining against the canonical ‘(dataset\_name, age)’ sample index, producing a data frame whose rows align with the community matrix and abiotic data used for model fitting.

## Usage

```
prepare_coords_for_fit(data_coords = NULL, data_sample_ids = NULL)
```

## Arguments

|                 |                                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| data_coords     | A data frame with ‘dataset_name’ as row names and columns ‘coord_long’ and ‘coord_lat’. |
| data_sample_ids | A data frame of valid ‘(dataset_name, age)’ pairs as returned by ‘align_sample_ids()’.  |

## Details

Coordinates are stored at dataset level but models require one row per sample. This function replicates each dataset’s coordinates across all its valid sample ages. The row-name format matches that of the community matrix and abiotic data frame produced by the respective preparation functions.

## Value

A data frame with row names in the format ‘"<dataset\_name>\_\_<age>"’ and columns ‘coord\_long’ and ‘coord\_lat’. Rows are sorted by ‘dataset\_name’ then ‘age’, matching the ordering of ‘data\_sample\_ids’.

**See Also**

[align\_sample\_ids()], [build\_jsdm\_fit\_input()]

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